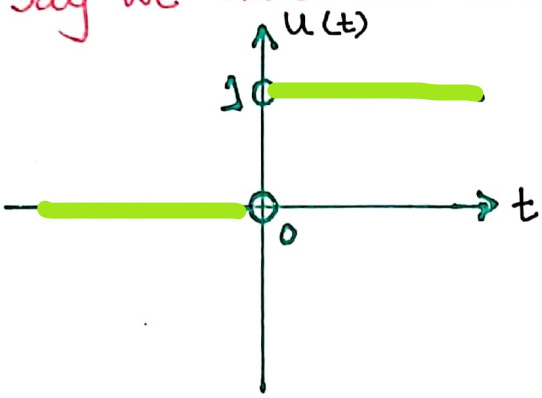
Laplace Transformation (Part B)

→ UNIT STEP FUNCTION or HEAVISIDE FUNCTION

Say we have the following signal:

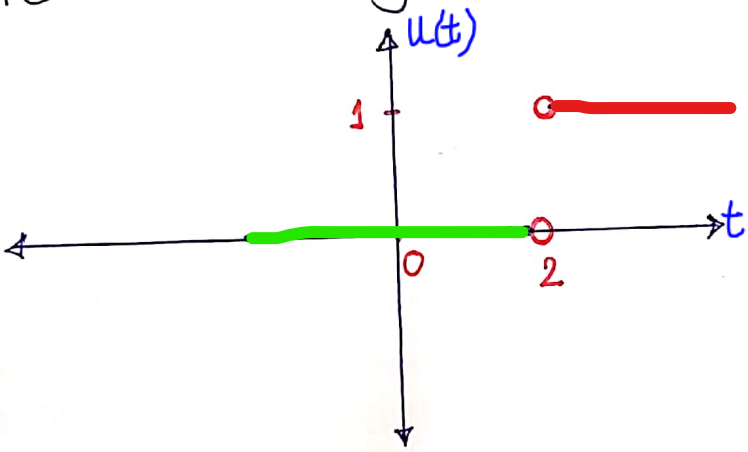


$$u(t) = \begin{cases} 0 & , t<0 \\ 1 & , t>0 \end{cases}$$

This is unit step function

$$u(t) \Rightarrow u(t-0)$$

Write the following graph in terms of function:



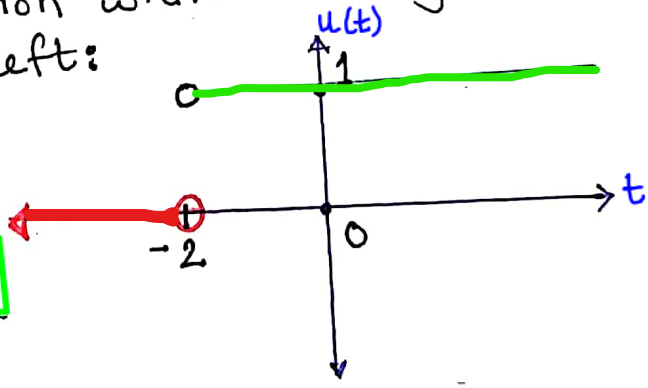
We are shifting towards right side

$$u(t-2) = \begin{cases} 0 & , \boxed{t<2} \\ 1 & , \boxed{t>2} \end{cases}$$

$$\begin{cases} t<2 \Rightarrow t-2<0 \\ t>2 \Rightarrow t-2>0 \end{cases}$$

Graph the following function while the signal is shifting two units towards left:

$$u(t+2) = \begin{cases} 0 & , \boxed{t<-2} \\ 1 & , \boxed{t>-2} \end{cases}$$



$$\begin{cases} t<-2 \Rightarrow t+2<0 \\ t>-2 \Rightarrow t+2>0 \end{cases}$$

Consider the following step function:

"a" is an arbitrary +ve number

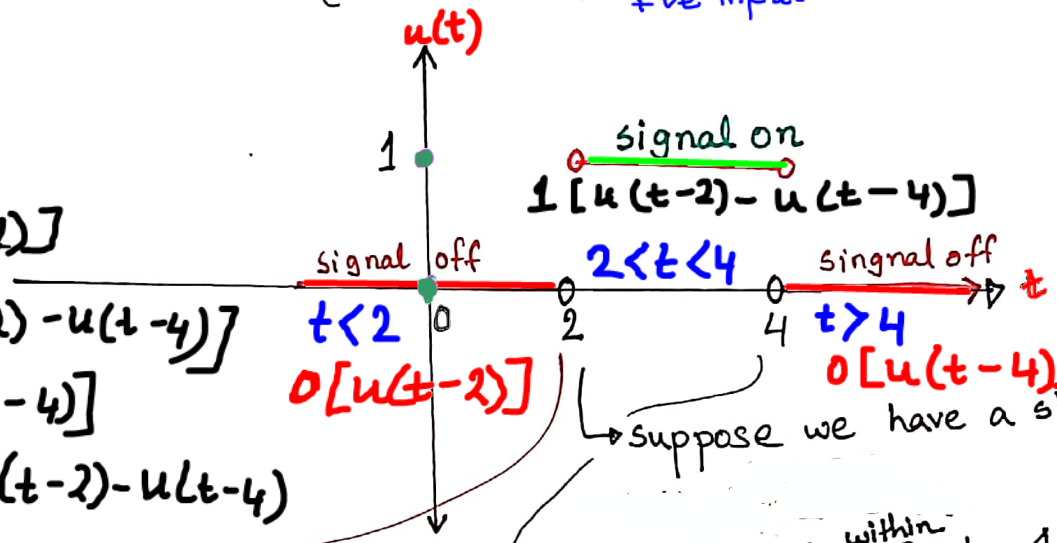
General structure of Unit Step Function:

$$u(t-a) = \begin{cases} 0, & \underbrace{t<a}_{\text{-ve input}} \\ 1 & \underbrace{t>a}_{\text{+ve input}} \end{cases} \qquad a \geq 0$$

$t-a<0 \Rightarrow t<a$

$t-a>0 \Rightarrow t>a$

$f(t)$
$= 0[u(t-2)]$
$+ 1[u(t-2)-u(t-4)]$
$+ 0[u(t-4)]$
$\therefore f(t) = u(t-2)-u(t-4)$



suppose we have a signal while t within 2 to 4 signal is on & when $t>4$, the signal is off.

Lets express this in terms of step function.

$$\boxed{f(t) = u(t-2) - u(t-4)}$$

The signal of the above graph can be presented mathematically in this way.

$t<2 \Rightarrow f(t) = \underbrace{u(t-2)}_{\text{-ve input}} - \underbrace{u(t-4)}_{\text{-ve input}} = 0 - 0 = 0$ (let $t=1$)

graphically output should be "0"

$2<t<4 \Rightarrow f(t) = \underbrace{u(t-2)}_{\text{+ve input}} - \underbrace{u(t-4)}_{\text{-ve input}} = 1 - 0 = 1$ (let $t=3$)

graphically output should be "1"

$t>2$ & $t<4 \Rightarrow u(t-2)$ is +ve
$t>2$ & $t<4 \Rightarrow u(t-4)$ is -ve

$\begin{cases} 0, & t<a \\ 1, & t>a \end{cases}$

$t>4 \Rightarrow f(t) = \underbrace{u(t-2)}_{\text{+ve input}} - \underbrace{u(t-4)}_{\text{+ve input}} = 1 - 1 = 0$ (let $t=5$)

graphically output should be "0"

It has to be "−" so the final answer is "0"

$t>4 \Rightarrow u(t-2)$ is +ve output = 1
$t>4 \Rightarrow u(t-4)$ is +ve $\therefore$ output = 1

If we change the function

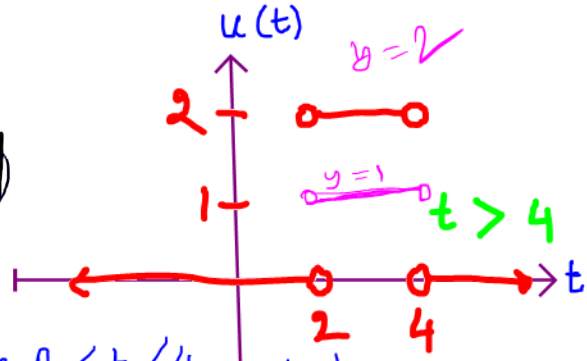
$$f(t) = 2\left[u(t-2) - u(t-4)\right]$$

+ve input +ve input

$t > 4$

$= 2(1 - 1) = 0$

let $t = 5$

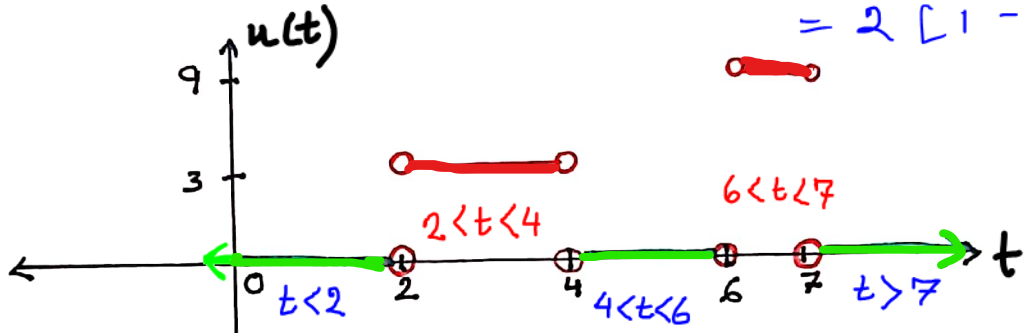


for: $2 < t < 4$ Let $t = 3$

$f(t) = 2[u(t-2) - u(t-4)]$

$= 2[1 - 0] = 2$ etc.

Consider another signal:



$f(t) = 0[u(t-2)] + 3[u(t-2) - u(t-4)] + 0[u(t-4) - u(t-6)] + 9[u(t-6) - u(t-7)] + 0[u(t-7)]$

$\therefore f(t) = 3[u(t-2) - u(t-4)] + 9[u(t-6) - u(t-7)]$

Represent the above function:

Step Function

$$f(t) = \begin{cases} 3, & 2 < t < 4 \\ 9, & 6 < t < 7 \\ 0, & o/w \end{cases}$$

$t < 2$ *, $t > 7$

$4 < t < 6$ **

Not given

$\therefore$ assuming over those intervals the signal is off.

Express this function in one line: (Unit step function)

$$f(t) = 3[u(t-2) - u(t-4)] + 9[u(t-6) - u(t-7)]$$

* $t < 2 \Rightarrow f(t) = 3[\underbrace{u(t-2)}_{-ve,\ \therefore\ "0"} - \underbrace{u(t-4)}_{-ve,\ "0"}] + 9[\underbrace{u(t-6)}_{-ve,\ "0"} - \underbrace{u(t-7)}_{-ve,\ "0"}] = 0$

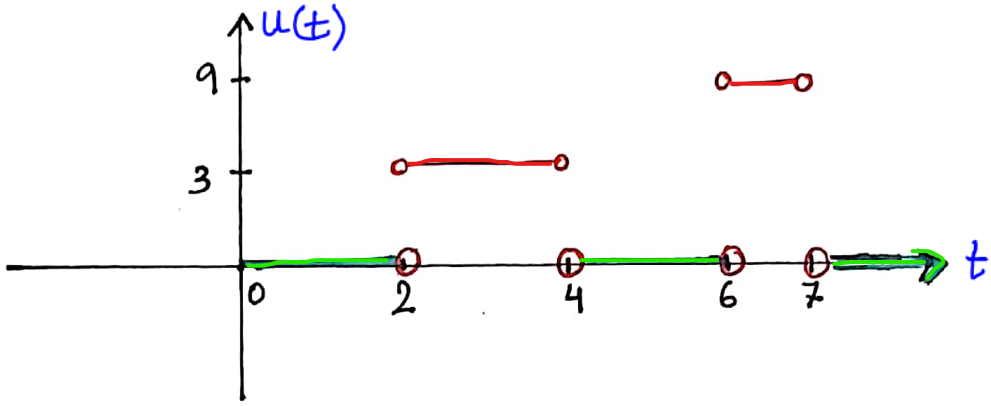
*let "t = 1"

$2 < t < 4 \Rightarrow f(t) = 3[(\underbrace{u(t-2)}_{+v,\ "1"} - \underbrace{u(t-4)}_{-ve,\ "0"}] + 9[\underbrace{u(t-6)}_{-ve,\ "0"} - \underbrace{u(t-7)}_{-ve,\ "0"}] = 3$

let "t = 3"

** $4 < t < 6 \Rightarrow f(t) = 3[\underbrace{u(t-2)}_{+ve,\ "1"} - \underbrace{u(t-4)}_{+ve,\ "1"}] + 9[\underbrace{u(t-6)}_{-ve,\ "0"} - \underbrace{u(t-7)}_{-ve,\ "0"}] = 0$

let "t = 5"



$$6<t<7 \Rightarrow f(t) = 3[u(\underbrace{t-2}_{\substack{+ve \\ \text{"1"}}}) - u(\underbrace{t-4}_{\substack{+ve \\ \text{"1"}}})] + 9[u(\underbrace{t-6}_{\substack{+ve \\ \text{"1"}}}) - u(\underbrace{t-7}_{\substack{-ve \\ \text{"0"}}})] = 9$$

let "t = 6.5"

Unit step function helps us to write the **step** function in one line.

Reading

<u>Example:</u> $f(t) = \begin{cases} t & ; 1<t<3 \\ \sin t & ; 6<t<7 \\ e^{2t} & ; t>7 \end{cases}$

$\rightarrow \because t<1 \rightarrow$ output $= 0$
$3<t<6 \rightarrow$ " $= 0$
$\Big\}$ $\because$ this intervals are missing in the information

Unit step function will be as follows:

$$f(t) = t[u(t-1) - u(t-3)] + \sin t[u(t-6) - u(t-7)] + e^{2t}[u(t-7)]$$

$$t<1 \Rightarrow f(t) = t[u(\underbrace{t-1}_{\substack{-ve \\ \text{"0"}}}) - u(\underbrace{t-3}_{\substack{-ve \\ \text{"0"}}})] + \sin t[u(\underbrace{t-6}_{\substack{-ve \\ \text{"0"}}}) - u(\underbrace{t-7}_{\substack{-ve \\ \text{"0"}}})] + e^{2t}(u(\underbrace{t-7}_{\substack{-ve \\ \text{"0"}}}))$$

$$= 0$$

$$1<t<3 \Rightarrow f(t) = t[u(\underbrace{t-1}_{\substack{+ve \\ \text{"1"}}}) - u(\underbrace{t-3}_{\substack{-ve \\ \text{"0"}}})] + \sin t[u(\underbrace{t-6}_{\substack{-ve \\ \text{"0"}}}) - u(\underbrace{t-7}_{\substack{-ve \\ \text{"0"}}})] + e^{2t}[u(\underbrace{t-7}_{\substack{-ve \\ \text{"0"}}})]$$

$$= t$$

$$6<t<7 \Rightarrow f(t) = t\,[\underbrace{u(t-1)}_{+ve\ \text{"}1\text{"}} - \underbrace{u(t-3)}_{+ve\ \text{"}1\text{"}}] + \sin t\,[\underbrace{u(t-6)}_{+ve\ \text{"}1\text{"}} - \underbrace{u(t-7)}_{-ve\ \text{"}0\text{"}}] + e^{2t}\,[\underbrace{u(t-7)}_{-ve\ \text{"}0\text{"}}]$$

$$= t\,[1-1] + \sin t\,[1-0] + e^{2t}\,[0]$$

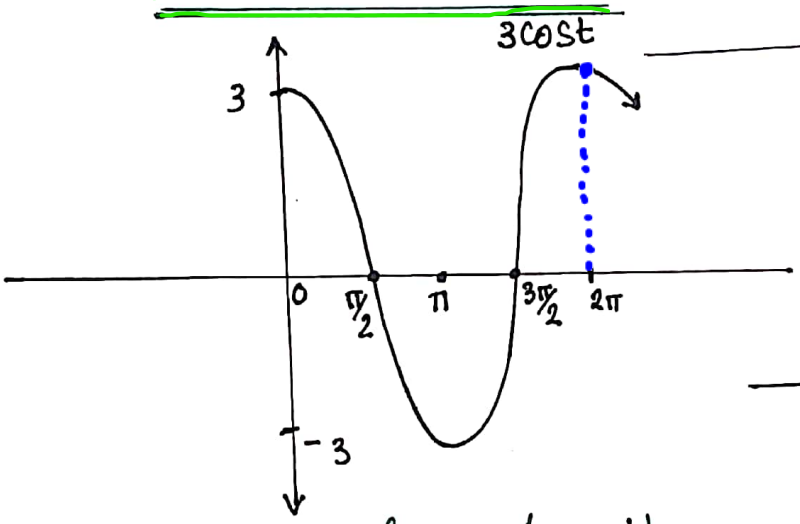
$$= \sin t$$

$$t>7 \Rightarrow f(t) = t\,[\underbrace{u(t-1)}_{+ve} - \underbrace{u(t-3)}_{+ve}] + \sin t\,[\underbrace{u(t-6)}_{+ve} - \underbrace{u(t-7)}_{+ve}] + e^{2t}[\underbrace{u(t-7)}_{+ve}]$$

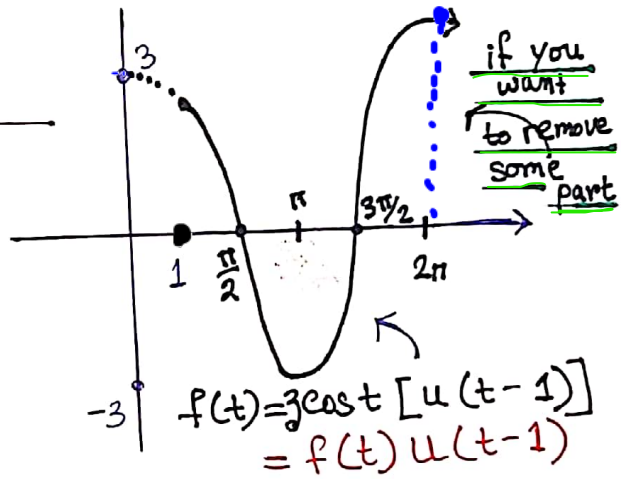
$$= t\,[1-1] + \sin t\,[1-1] + e^{2t}\,(1)$$

$$= e^{2t}$$

Omit these Graphs → TRANSLATIONS



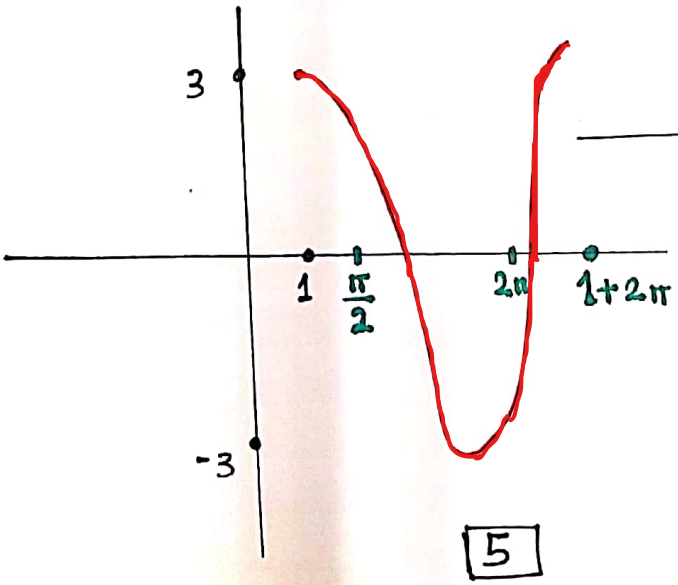
→ $f(t) = 3\cos t$



if you want to remove some part

$f(t) = 3\cos t\,[u(t-1)]$
$= f(t)\,u(t-1)$

shift it forward by 1 unit



→ $f(t) = 3\cos(t-1)\,[u(t-1)]$
$= f(t-1)\,u(t-1)$
Translation

Example $\boxed{1}$

$$f(t) = \begin{cases} 0 & ; 0<t<1 \\ t-1 & ; 1<t<2 \\ t+1 & ; t>2 \end{cases}$$

Find the Laplace transformation of the above function

Unit step function:

$$f(t) = 0[u(t-0)-u(t-1)] + (t-1)[u(t-1)-u(t-2)] + (t+1)[u(t-2)]$$

$$= tu(t-1) - tu(t-2) - u(t-1) + u(t-2) + tu(t-2) + u(t-2)$$

$$= tu(t-1) - u(t-1) + 2u(t-2)$$

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{tu(t-1)\} - \mathcal{L}\{u(t-1)\} + 2\mathcal{L}\{u(t-2)\}$$

take laplace on both sides

1st Translation Theorem:

$$\mathcal{L}\{u(t-a)f(t)\} = e^{-sa}\mathcal{L}\{f(t+a)\}$$

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{t\,u(t-1)\} - \mathcal{L}\{1u(t-1)\} + 2\mathcal{L}\{1u(t-2)\}$$

$a=1$ $\quad$ $f(t)=1$ $\quad$ $a=1$ $\quad$ $f(t)=1$ $\quad$ $a=2$

$f(t)=t$ $\quad$ $\therefore f(t+a) = f(t+1) = 1$ $\quad$ $\therefore f(t+a) = f(t+2) = 1$

$f(t+a) = f(t+1) = t+1$

$$\mathcal{L}\{f(t)\} = e^{-s(1)}\mathcal{L}\{f(t+1)\} - e^{-s(1)}\mathcal{L}\{f(t+1)\} - 2e^{-s(2)}\mathcal{L}\{f(t+2)\}$$

$$= e^{-s}\mathcal{L}\{t+1\} - e^{-s}\mathcal{L}\{1\} - 2e^{-2s}\mathcal{L}\{1\}$$

$$= e^{-s}\left(\frac{1}{s^2}+\frac{1}{s}\right) - e^{-s}\cdot\frac{1}{s} + 2e^{-2s}\frac{1}{s}$$

Scanned with CamScanner

$$= e^{-s}\frac{1}{s^2} + e^{-s}\frac{1}{s} - e^{-s}\frac{1}{s} + 2e^{-2s}\frac{1}{s}$$

$$= \frac{1}{s^2}e^{-s} + \frac{2}{s}e^{-2s}$$

<u>Example</u> $\boxed{2}$ Find the Laplace Transformation of

$$f(t) = \begin{cases} 2; & 0<t<3 \\ t^2; & 3<t<5 \\ t+1; & t>5 \end{cases}$$

$$\mathcal{L}\{u(t-a)f(t)\} = e^{-sa}\mathcal{L}\{f(t+a)\}$$

Unit step function:

$$f(t) = 2[u(t-0) - u(t-3)] + t^2[u(t-3) - u(t-5)] + (t+1)[u(t-5)]$$

$$= 2u(t-0) - 2u(t-3) + t^2u(t-3) - t^2u(t-5)$$
$$+ t(u(t-5)) + 1u(t-5)$$

take Laplace on both sides:

$$\mathcal{L}\{f(t)\} = 2\mathcal{L}\{1u(t-\overset{a}{0})\} - 2\mathcal{L}\{1u(t-\overset{a}{3})\} + \mathcal{L}\{t^2u(t-\overset{a}{3})\}$$

$f(t)=1$, $f(t+0)=1$; $f(t)=1$, $f(t+3)=1$; $f(t)=t^2$, $f(t+3)=(t+3)^2$

$$- \mathcal{L}\{t^2u(t-\overset{a}{5})\} + \mathcal{L}\{tu(t-\overset{a}{5})\} + \mathcal{L}\{1u(t-\overset{a}{5})\}$$

$f(t)=t^2$, $f(t+5)=(t+5)^2$; $f(t)=t$, $f(t+5)=t+5$; $f(t)=1$, $f(t+5)=1$

$$\mathcal{L}\{u(t-a)f(t)\} = e^{-sa}\mathcal{L}\{f(t+a)\}$$

$$= 2e^{-s(0)}\mathcal{L}\{f(t+0)\} - 2e^{-s(3)}\mathcal{L}\{f(t+3)\} + e^{-s(3)}\mathcal{L}\{f(t+3)\}$$
$$- e^{-s(5)}\mathcal{L}\{f(t+5)\} + e^{-s(5)}\mathcal{L}\{f(t+5)\} + e^{-s(5)}\mathcal{L}\{f(t+5)\}$$

$$= 2.1.\mathcal{L}\{1\} - 2e^{-3s}\mathcal{L}\{1\} + e^{-3s}\mathcal{L}\{t+3\}^2$$
$$- e^{-5s}\mathcal{L}\{(t+5)^2\} + e^{-5s}\mathcal{L}\{t+5\} + e^{-5s}\mathcal{L}\{1\}$$

$$= 2.\frac{1}{s} - 2e^{-3s}\frac{1}{s} + e^{-3s}\mathcal{L}\{t^2+6t+9\}$$
$$+ e^{-5s}\mathcal{L}\{t^2+10t+25\} + e^{-5s}\left(\frac{1}{s^2}+\frac{5}{s}\right) + e^{-5s}\cdot\frac{1}{s}$$

Scanned with CamScanner

$$= \frac{2}{s} - \frac{2}{s} e^{-3s} + e^{-3s}\left(\frac{2b}{s^{2+1}} + 6 \cdot \frac{1}{s^2} + \frac{9}{s}\right)$$

$$+ e^{-5s}\left(\frac{2b}{s^{2+1}} + 10 \cdot \frac{1}{s^2} + 25 \frac{1}{s}\right) + e^{-5s}\left(\frac{1}{s^2} + \frac{5}{s}\right) + \frac{e^{-5s}}{s}$$

$$= \frac{2}{s} - \frac{2e^{-3s}}{s} + \frac{2e^{-3s}}{s^3} + \frac{6e^{-3s}}{s^2} + \frac{9e^{-3s}}{s}$$

$$+ \frac{2e^{-5s}}{s^3} + \frac{10e^{-5s}}{s^2} + \frac{25e^{-5s}}{s} + \frac{e^{-5s}}{s^2} + \frac{5e^{-5s}}{s} + \frac{e^{-5s}}{s}$$

$$= \frac{2}{s} + \frac{7e^{-3s}}{s} + \frac{2e^{-3s}}{s^3} + \frac{6e^{-3s}}{s^2}$$

$$+ \frac{2e^{-5s}}{s^3} + \frac{11e^{-5s}}{s^2} + \frac{31e^{-5s}}{s}$$

Scanned with CamScanner